

AYMANE AIT DADS

AI Machine Learning Engineer (Intern) | End-to-End ML Pipelines · Deep Learning · MLOps

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PROFESSIONAL SUMMARY

Built and deployed a production-grade ML pipeline on 100K+ fiber-optic network records at Orange Maroc, automating feature engineering and model evaluation to reduce manual analysis time by ~60%. Ranked top 10% (0.80/1.0) in the NVIDIA Nemotron Reasoning Kaggle Competition (\$106K+ prize pool) by fine-tuning a 30B Mamba-Transformer MoE with LoRA and executing 20+ single-variable ablations to guide training decisions. Core competencies span end-to-end ML pipeline development, deep learning with PyTorch and Hugging Face Transformers, computer vision, NLP, and MLOps including CUDA debugging, mixed-precision training, and cross-validated model evaluation. Available for a US-based internship; eligible to work in the USA under a student visa (F-1/OPT sponsorship may be required).

CORE COMPETENCIES

End-to-End ML Pipelines | Deep Learning | MLOps | Feature Engineering | Model Evaluation | Computer Vision | NLP | Cross-Functional Collaboration

WORK EXPERIENCE

Orange Maroc

Jul 2024 - Aug 2024

Data Science Intern - Casablanca, Morocco

- Designed an end-to-end ML pipeline combining Random Forest classification and K-Means clustering on **100K+ fiber-optic network records** (upload speed, latency, jitter) to classify quality across thousands of nodes.
- Engineered a clustering pipeline that mapped customer performance profiles to **5 commercial service tiers**, with output adopted directly by the network optimization team for infrastructure decisions.
- Reduced manual analysis time by **~60%** through automated feature engineering, cross-validated model evaluation, and systematic hyperparameter tuning using Scikit-learn and Pandas.
- Delivered stakeholder presentations translating model outputs into actionable maintenance recommendations, supported by **Power BI** dashboards for non-technical audiences.

PROJECTS

NVIDIA Nemotron Reasoning Challenge

Kaggle Competition · In Progress June 2026 · Top 10% Public Leaderboard

- Achieved **top 10% public leaderboard (0.80/1.0)** in the \$106K+ prize pool competition by fine-tuning a 30B Mamba-Transformer MoE with LoRA ($r=32$, ~888M trainable parameters) via SFTTrainer on 7,828 chain-of-thought logic examples.
- Resolved **5 Blackwell GPU (sm_120) incompatibilities** through targeted monkey-patches and PyTorch fallbacks, and ran 20+ single-variable ablations confirming completion-only loss and packing strategy as decisive training variables.

PyTorch · Hugging Face Transformers · Unsloth · PEFT · vLLM · Triton

AI-Generated Image Detection

EURECOM ImSecu + NTIRE 2026 @ CVPR · 2025 · Ranked 1st in Class

- Ranked **1st in EURECOM class private Kaggle leaderboard (0.791 AUC)** by fine-tuning CLIP ViT-L/14 (428M params) with a custom MLP head, dual learning rates, and a 3-model ensemble with 10-view TTA across 250K images from 25+ generator types.
- Submitted to the **NTIRE 2026 @ CVPR CodaBench** international benchmark; ablation studies revealed 1-epoch training (0.793 AUC) outperformed 4-epoch (0.767 AUC), confirming out-of-distribution generalization as the critical optimization target.

PyTorch · OpenCLIP · Hugging Face Transformers · Kaggle GPU · FP16

EDUCATION

Engineering Degree in Data Science (Master-level) - EURECOM

Sep 2023 - Present

Relevant coursework: Machine Learning, Deep Learning, Computer Vision, NLP, Statistics, Cloud Computing, Image Security, Web Applications

CPGE - Mathematics & Physics - Ibn Timiya

2021 - 2023

SKILLS

ML Frameworks & Engineering: PyTorch | Scikit-learn | Hugging Face Transformers | TensorFlow | PEFT | Unsloth | TRL

AI Domains & Techniques: Computer Vision | NLP | LoRA Fine-tuning | Feature Engineering | Clustering | Ensemble Methods | Chain-of-Thought

MLOps, Data & Tools: Python | SQL | Git | FP16 Mixed Precision | CUDA Debugging | Pandas | Power BI | Docker